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Variation on a Trivialist Argument of Paul Kabay

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Abstract Impossible worlds are regarded with understandable suspicion by most philosophers. Here we are concerned with a modal argument which might seem to show that acknowledging their existence, or more particularly, the existence of some hypothetical (we do not say "possible") world in which everything was the case, would have drastic effects, forcing us to conclude that everything is indeed the case—and not just in the hypothesized world in question. The argument is inspired by a metaphysical (rather than modal-logical) argument of Paul Kabay's which would have us accept this unpalatable conclusion, though its details bear a closer resemblance to a line of thought developed by JC Beall, in response to which Graham Priest has made some philosophical moves which are echoed in our diagnosis of what goes wrong with the present modal argument. Logical points of some interest independent of the main issue arise along the way.

Keywords Impossible worlds · Modal logic · Trivialism

1 Introduction

Kabay (2009) gives a sympathetic airing to an argument whose conclusion is that the actual world is what Kabay calls a trivial world, understanding a world w as trivial when everything is true in w, or perhaps more explicitly when every statement (or every _, where the blank is filled by a specification of one's favoured class of candidate truth-bearers) is true in w. The starting point in this argument is the claim that there is a trivial world, where 'world' is not understood to mean 'possible world' but to embrace also impossible worlds—since impossible is what, on the most natural view,

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a trivial world would be.[1] After various intermediate steps involving details about the mereology of worlds which we need not go into here, the argument concludes that the actual world is itself trivial, which amounts to the defining claim of the dialectically unusual position known as trivialism. (See Priest (2000); the position is mentioned, though not under that name, at p. 331 of Mortensen (1989) along with earlier work by Putnam.) In fact, Kabay takes it that there is a unique trivial world—a feature replicated in the model-theoretic treatment below—and his argument concludes that the actual world is identical with this unique trivial world.

The present variation on that line of thought is concerned only to establish the conditional that *if* there is a trivial world, *then* the actual world is trivial, rather than (like Kabay's) with going further and asserting the antecedent, and hence the consequent, of this conditional. Further, though this talk of worlds will be in place in discussing the semantics of the language in play in the current argument, the argument itself will be cast in a broadly modal object language, and will bear a closer resemblance to very similar considerations aired in Beall (2006, 2009)—Sect. 5 below—than to Kabay's discussion.

The core idea of the argument under consideration here can be conveyed with the aid of the language of propositional modal logic (with the boolean connectives and the modal operators $\Box$ and $\Diamond$, with the usual stock of propositional variables, of which we take p and q to be two) enhanced by propositional quantifiers. We will then reformulate the argument so that this last—potentially complicating—enhancement drops away.

(1)	$\Diamond\forall p\,.\,p$	Modalized trivialism hypothesis
(2)	$\Diamond\Box\forall q\,.\,q$	From (1), as explained below
(3)	$\forall q\,.\,q$	From (2) using $\Diamond\Box A \rightarrow A$.

Line (1) here is our modal way of saying that there is a trivial world, though one may think that "$\Diamond$", being commonly read "it is possible that", is a rather perverse notation to use for this purpose. With this consideration in mind, in the following section we will write, not "$\Diamond A$", to say that A is true in some world, but "$\Diamond^+ A$" with that reading in mind. So $\Diamond$ there will be taken as existentially quantifying over *possible* worlds while $\Diamond^+$ quantifies existentially (and $\Box^+$ universally) over *all* worlds, which we will take to comprise the possible worlds and a single impossible world—the trivial world, at which everything is true. But above we have written $\Diamond$ and $\Box$ without the "+" superscripts indicating the wider range. Line (2) could equally well have been written with "$\Box\forall p\,.\,p$" in place of "$\Box\forall q\,.\,q$"; the variable has been changed only to make the

[1] Such objects have found their way into model-theoretic semantics as technical adjuncts—the 'absurd' world introduced under the name "λ" on p. 103 of Stalnaker (1968), the 'sick' (or 'exploding') points of Veldman (1976)—and their general claim to be taken seriously has been much debated in the literature. See Berto (2009) and references there cited, to which we may add Cresswell (1975) and the special issue (#4) of the *Notre Dame Journal of Formal Logic* **38** (1997), edited by Graham Priest. As this literature reveals, there are two aspects of impossibility that may arise: a world may on the one hand be described as verifying too much for the description to represent a possibility at all, and, on the other, it may be described as verifying too little for the description to qualify as the complete description of a single possible world. Impossibility by commission and by omission, respectively, we may briefly say to summarise this contrast. In the case of the trivial world(s) in play here, only the former feature is on display. (The latter is touched on briefly in note 11.)

justification on the right-hand side less potentially confusing. Either way the content is the same: all worlds are trivial. The justification here (as promised in the annotation given for line (2))—is that "$\Box\forall q \centerdot q$" follows from "$\forall p \centerdot p$" by elimination of the universal propositional quantifier (instantiating "p" as "$\Box\forall q \centerdot q$"), and therefore, we pass to line (2) exploiting the *monotone* or 'entailment preserving' property of $\Diamond$: if B follows from A, then $\Diamond B$ follows from $\Diamond A$. Since we are thinking of "$\Diamond$" as an existential quantifier over worlds, this is evidently a correct principle for "$\Diamond$". Finally, line (3) is justified by citing another modal principle, the dual form of the Brouwerian principle (often called B), a well-known principle available in the normal modal logic S5, which is sound w.r.t. any class of models in which $\Box$ and $\Diamond$ are interpreted—as here—by means of universal and existential quantification, respectively, over the totality of points in the model (the points here being the possible worlds together with the trivial world).[2]

2 Restating the Argument

Dropping the propositional quantifiers from the argument as laid out in the preceding section and instead treating the hypothesis of modalized trivialism as a candidate axiom in a logic with Uniform Substitution (of arbitrary formulas for propositional variables), we get the following, in which, in accordance with the suggestion just made, we write $\Box^+$ and $\Diamond^+$ in place of $\Box$ and $\Diamond$:

$(1)^+$	$\Diamond^+ p$	Modalized trivialism axiom
$(2)^+$	$\Diamond^+\Box^+ q$	From $(1)^+$ by Uniform Substitution ($\Box^+ q$ for p)
$(3)^+$	q	From $(2)^+$ using $\Diamond^+\Box^+ A \to A$.

The transition from line $(2)^+$ to line $(3)^+$ is justified—so the proponent of this line of thought would have it—as before (i.e., in passing from (2) to (3)): the B schema is available because we are using the universal accessibility relation on the set of worlds (comprising the possible worlds together with the trivial world), and this relation is symmetric.

A propos of the transition from $(1)^+$ to $(2)^+$, again, as in the original version of the argument, there is no need to shift from "p" to "q" here. Since we are assuming Uniform Substitution available as a rule, every formula is provable in the envisaged logic—a property sometimes called inconsistency and sometimes called triviality.[3]

[2] As the reader will recognise, it is because the (tacit) universal accessibility relation in play here is *symmetric* that B is a valid principle, which may occasion a provisional response to the above derivation along the following line: we need to retreat to a weaker modal logic than S5, one determined by a class of frames not all of which have a symmetric accessibility relation. Such a response would not be plain sailing, however, since if the Kripke semantics is taken seriously, the accessibility relation R interpreting $\Box$ will still have a converse, R^{-1}, and we should be able to introduce a modal operator, $\Box^{-1}$, say, understood quantifying over worlds accessible by the relation R^{-1}, in which case we replace the formula at line (2) with $\Diamond\Box^{-1}\forall q \centerdot q$, and the implication justifying the transition to line (3), $\Diamond\Box^{-1} A \to A$ cannot now be contested. (This line of thought, with a different application, can be found in Humberstone 1983.)

[3] Since we are not departing from classical truth-functional logic in respect of the behaviour of our non-modal logical vocabulary, negation inconsistency and absolute inconsistency—alias triviality—coincide. (The 'absolute' terminology can be found in, for example, Church 1956, p. 108.)

Before we bid farewell to the propositional quantifiers forever, we should note that the new $(1)^+$–$(3)^+$ does not simply recapitulate the original (1)–(3) version, since the effect, put in terms of propositional quantifiers, of the propositional variables in a formula taken as provable is as bound by an outermost universal propositional quantifier. (Uniform Substitution does the work of an elimination rule for this quantifier.) This means that $(1)^+$ in effect has the force of $\forall p \,.\, \Diamond^+ p$, rather than of the old line (1), with its diamond suitably marked with a "+": $\Diamond^+ \forall p \,.\, p$. The old line (1) is a hypothesis of the possible$^+$ truth of trivialism (its truth in some world that is: we use "could$^+$" similarly), while the new line $(1)^+$ is rather different, telling us not that *everything* could$^+$ be true, but only—an apparently weaker claim—that *anything* could$^+$ be true. But the apparent weakness is apparent only, as we see, since if anything could$^+$ be true, one of the things that could$^+$ be true is that everything is true.[4] This underlies the reconstructability of the original argument in a form from which the propositional quantifiers are absent. (Note that in the original (1)–(3) argument, if (1) had been replaced with what might have seemed to be the weaker assumption $\forall p \,.\, \Diamond p$, line (2) would follow by $\forall$-Elimination.)

Now of course the claim that anything whatever (or indeed everything) could$^+$ be true is not the claim that anything whatever could be true, but since it has been explained in terms of the existence of worlds understood so as to encompass not only possible but also impossible worlds, one might take the argument, either in the original (1)–(3) form or in its $(1)^+$–$(3)^+$ incarnation, as underlining the non-existence of impossible worlds, which (Lewis 1986, p. 7, note 3) told us long ago were something whose postulation involved us in contradicting ourselves. Thus one might react by transferring, in the light of (1)–(3) or $(1)^+$–$(3)^+$, one's rejection of trivialism to a rejection of the first step of either of these arguments. Such a Modus Tollens reaction, no less than Kabay's own Modus Ponens reaction, assumes that the arguments are valid, however, and to assess that assumption, we should look more closely.

3 Semantic Apparatus

We use the notation "t" to suggest "trivial world", in introducing models as triples $\langle W, t, V\rangle$ in which $t \in W$ and V assigns subsets of W to propositional variables. We want to engineer the definition of truth so that every formula comes out true at the element t in any such model. One way to do this—though by itself it would not be sufficient— would be at this early stage to stipulate that $t \in V(p_i)$ for each p_i. (Here we are thinking of the propositional variables as $p_1, \ldots, p_n, \ldots$—the first two of which we continue to abbreviate to "p" and "q" when convenient—and of the formulas as generated from these by application of the boolean connectives, $\neg$, $\wedge$, ..., as well as $\Box^+$ and, for good measure $\Box$ itself, and we shall also want their duals $\Diamond^+$ and $\Diamond$—though the "duals" terminology will turn out—see Sect. 4—to be problematic.)

[4] In the logical setting in which we are working, instead of saying that one of the things that could$^+$ be true is that everything is true, we could simply say that one of the things that could$^+$ be true is $p \wedge \neg p$, or, more simply still: one such thing is $\perp$. (The latter is officially introduced in Sect. 4.)

With that stipulation in force, we can use the following standard basis clause for the definition of truth of a formula A at a point x in a model $\mathcal{M} = \langle W, t, V \rangle$, where it is assumed that $x \in W$:

$$\mathcal{M} \models_x p_i \text{ iff } x \in V(p_i).$$

Call that the first strategy. An alternative—call it the second strategy—which also results in its being the case that for any such model $\mathcal{M}$, we have $\mathcal{M} \models_t p_i$ for all p_i would be to impose no condition concerned t on V, but to override whatever V says when it comes to truth at t. In other words, one would say:

$$\mathcal{M} \models_x p_i \text{ iff } (i)\ x = t \text{ or } (ii)\ x \neq t \quad \text{and } x \in V(p_i).$$

There is not much to choose between the first strategy and the second strategy here, or even when it comes to considering $\wedge$ and $\vee$ (just supposing we decide to take both as primitive). We can continue the first strategy with standard clauses:

$$\mathcal{M} \models_x A \wedge B \text{ iff } \mathcal{M} \models_x A \quad \text{and} \mathcal{M} \models_x B,$$

and similarly in the case of $\vee$. Alternatively, following the bipartite division in the atomic case on the second strategy, we could split cases and say:

$$\mathcal{M} \models_x A \wedge B \text{ iff } (i) x = t \text{ or } (ii) x \neq t \text{ and both } \mathcal{M} \models_x A \text{ and } \mathcal{M} \models_x B,$$

and similarly for $\vee$.[5] It makes no difference—in the absence of further logical vocabulary—which approach is followed, since an induction on the complexity of formulas constructed from propositional variables by means of conjunction and disjunction guarantees that all are true at a point (such as, in particular, the point t) at which all the propositional variables are true. When it comes to negation, however, the two approaches are not equally good, and we need the bipartite treatment of the second strategy (or the alternative non-uniform formulation as in Note 5):

$$\mathcal{M} \models_x \neg A \text{ iff } (i) x = t \text{ or } (ii) x \neq t \text{ and } \mathcal{M} \not\models_x A.$$

We can't just make do with the second conjunct of (ii), having dropped (i), since this will turn formulas A true at t into formulas $\neg A$ not true at t, and we wanted all formulas to end up true at t (in $\mathcal{M}$). A more straightforward presentation of the second strategy, which we will take as definitive, would deal with the two cases separately, and say that for any model $\mathcal{M} = \langle W, t, V \rangle$ as above (and with or without the condition that $t \in V(p_i)$ for all p_i) we have, for all formulas A, B, and all $x \in W$.

[5] In such clauses, we can without loss drop the "$x \neq t$ and" part, which is retained here simply as a reminder of 'non-uniform' reformulations—i.e., giving separate necessary and sufficient conditions for the truth of a formula at x depending on whether or not $x = t$—along the lines of (for the present case): "If $x = t$ then $\mathcal{M} \models_x A \wedge B$, and if $x \neq t$ then $\mathcal{M} \models_x A \wedge B$ iff $\mathcal{M} \models_x A$ and $\mathcal{M} \models_x B$". This uniformity issue comes up again in Sect. 5.

If $x = t$:	If $x \neq t$:
$\mathcal{M} \models_x A$	$\mathcal{M} \models_x A \wedge B$ iff $\mathcal{M} \models_x A$ and $\mathcal{M} \models_x B$; $\mathcal{M} \models_x A \to B$ iff $\mathcal{M} \not\models_x A$ or $\mathcal{M} \models_x B$; $\mathcal{M} \models_x \neg A$ iff $\mathcal{M} \not\models_x A$; $\mathcal{M} \models_x \Box A$ iff for all $y \in W \setminus \{t\}$, $\mathcal{M} \models_y A$; $\mathcal{M} \models_x \Box^+ A$ iff for all $y \in W$, $\mathcal{M} \models_y A$.

Here in the second column the case of $\to$ (material implication) has also been treated; we could have included $\vee$ as well ($\mathcal{M} \models_x A \vee B$ iff $\mathcal{M} \models_x A$ or $\mathcal{M} \models_x B$), and for good measure we could attend to $\Diamond$ and $\Diamond^+$ too, though for the moment let us just consider the logic of the non-modal connectives on this account, with a view to asking about its classicality. There is always the question of whether to consider the logic as (1) a set of formulas, those true at all points in all of the models when the above truth-definition is in force, or (2) as a consequence relation holding between a set of formulas and a given individual formula when at any point in one of these models at which all formulas in the set are true, the given formula is true, or (3) as a generalized (or 'multiple-conclusion') consequence relation holding between two sets of formulas when at any point in one of these models at which all formulas in the first set are true, at least one formula in the second set is true. When attention is restricted to non-modal formulas, the logic we get as a consequence relation (conception (2)), and therefore, also as a set of formulas (conception (1)), taking the consequences of the empty set, is just classical propositional logic,[6] while when taken in accordance with conception (3), as a generalized consequence relation—$\Vdash$, say—we have something weaker than the corresponding classical logic since for the present account it is no longer the case that $A, \neg A \Vdash \varnothing$.[7] Recall from Note 3 that we do *not* have a paraconsistent logic on our hands here, in the sense that we *do* have $A, \neg A \Vdash B$ for all formulas A, B, as follows from the fact that the generalized consequence relation agrees with the consequence relation in the one-formula-on-the-right case, and the already noted classicality of the

[6] This is a reworking of the familiar fact that this is the consequence relation determined by the class of boolean valuations—bivalent truth-value assignments respecting the standard truth-tables, that is—is also determined by the union of this class with the single (non-boolean) valuation which assigns the value True to every formula. (This last valuation corresponds to the presence of t in our models.) This is a special case of the fact that the set of valuations consistent with a consequence relation is closed under what in Humberstone (1996) is called the operation of conjunctive combination, as applied to the empty set of valuations.

[7] This is a convenient abbreviated way of writing "$\{A, \neg A\} \Vdash \varnothing$". Here a rather tricky question arises as to whether, if logics are identified with generalized consequence relations, "the trivialist *is a classical logician*", as Priest (2000), p. 189 writes (his emphasis). I have just said that the empty set is not a generalized consequence of a premiss-set containing A and $\neg A$ on the current semantics, tailored not to trivialism but modalized trivialism (of the "everything is possible$^+$" form), because the trivial world does not verify some formula in the empty set—there being no such formulas—even though it does verify the premisses. But the modal trivialist who happens also to be a trivialist *tout court* will assent to all claims, including the claim that the (generalized) argument just described, from A and $\neg A$ to $\varnothing$, is valid (and also, disconcertingly, to the claim that it is not valid). Here we assume that the trivialist possesses the concepts deployed in these claims. The necessity for this assumption shows, incidentally. that we should not infer, from the fact that the trivialist believes that every proposition is true, that the trivialist believes every proposition. But the disconcerting feature just mentioned is enough to make for the awkwardness (touched on in Priest 2000 and expanded on in yet-to-be published work of Kabay) of trying to disagree with a trivialist on any issue involving concepts you both grasp, since the trivialist will agree with any claim you make on the issue.

latter.[8] For the remainder of the discussion we revert to conception (1): logics as sets of formulas.

The modal vocabulary needs to be discussed too, since it was this vocabulary that was deployed so conspicuously in the argument (1)–(3) with which we began, and in its subsequent restatement as $(1)^{+}$–$(3)^{+}$ in Sect. 2. We provide this discussion in the course of treating a puzzle, after which we return to diagnosing what goes wrong with these arguments.

4 Resolving a Puzzle

The observation from Note 6 about adding the constant-true valuation to a set of valuations not making a difference to the logic determined when logics are taken as consequence relations or just as sets of formulas has the following repercussion for the current models. Suppose we are deciding between two notions of validity for formulas. The first, employed above though not under this name, counts a formula as valid just in case it is true at every $x \in W$ in every $\mathcal{M} = \langle W, t, V\rangle$. The second notion is inspired by the observation that validity is sometimes defined, for frames in modal logics in which some points are deemed normal and others non-normal, to consist in being true at all the normal points, rather than at all points whatever, in every model (on the frame in question). The notion of validity inspired by this would note that t looks rather like some of the non-normal points in the models just alluded to and accordingly requires for a formula's validity that it should be true at every $x \in W \smallsetminus \{t\}$ in every $\mathcal{M} = \langle W, t, V\rangle$. Now a version of the observation alluded to at the start of this paragraph gives us the conclusion: the two notions of validity just distinguished are in fact co-extensive, since of course validity in the first sense implies validity in the second, which removes from the models one point (namely t) at which a counterexample (to the hypothesis that a formula is valid) might present itself; but on second thoughts we see that since t verifies all formulas it wasn't the locus for a possible counterexample (to that hypothesis) after all, and validity in the second sense implies validity in the first.[9]

To get to the promised puzzle, we apply this same reasoning not in connection with validity but in connection with the truth of $\Box$ and $\Box^{+}$ formulas. Take any model

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[8] 'Classicality' in respect of the non-modal formulas, that is, the notion having no settled meaning when modal formulas are also considered. (In the sense of 'classical' for modal logics used in Segerberg (1971)—which amounts to having the replacement property described in Note 10 below—and therefore, also in the weaker sense of this terminology in Segerberg (1982), all logics considered here are classical.) Mortensen (1989) defines *weak paraconsistentism* as the view that "no contradictions are true, but that inconsistent 'worlds' need to be added to one's model." Presumably that should be "models"; the second part of this claim is certainly part of the current package. While the current semantics provides every model with a trivial world, in the Appendix we consider a more generous approach which allows but does not insist on such a world in the models. It is essential that they be allowed for, in the sense that if we restricted attention to models in which there were no such worlds, the set of valid formulas would be different.

[9] By the reasoning in the previous section, this claim of co-extensiveness for the two notions of validity would have to be retracted if we were considering the validity of arguments or sequents in a multiple-conclusion framework, since the validity status of arguments of the form "$A, \neg A \therefore \varnothing$" would be affected by the choice between them.

$\mathcal{M} = \langle W, t, V \rangle$ and any $x \in W$. If $x = t$, then for any formula A, $\mathcal{M} \models_x \Box A$ iff $\mathcal{M} \models_x \Box^+ A$, since $\Box A$ and $\Box^+ A$ are both true (along with all other formulas) at t in $\mathcal{M}$. Next suppose that $x \neq t$. Then evidently

$$\mathcal{M} \models_x \Box^+ A \Rightarrow \mathcal{M} \models_x \Box A,$$

since for $\mathcal{M} \models_x \Box^+ A$ we need A true (in $\mathcal{M}$) throughout W, which implies that A is true throughout $W\{t\}$, which suffices for $\mathcal{M} \models_x \Box^+ A$. But also conversely, we have

$$\mathcal{M} \models_x \Box A \Rightarrow \mathcal{M} \models_x \Box^+ A,$$

since if $\mathcal{M} \models_x \Box A$ then A is true throughout $W \smallsetminus \{t\}$, and A like any other formula, is also true at t, making A true throughout W, which suffices for $\mathcal{M} \models_x \Box^+ A$.

We still haven't quite got to the puzzle yet—though it is interesting to see that the apparent distinction between necessity and necessity$^+$ has proved to be null and void. The puzzle is that the collapse of this distinction seems to threaten the distinction between possibility and possibility$^+$, whose significance was stressed in Sects. 1 and 2. Let us go slowly through what is generally a familiar derivation of the unpalatable direction ($\Diamond^+ A \to \Diamond A$) of the equivalence in question:

(i)	$\Box\neg A \to \Box^+\neg A$	Special case of $\Box \Rightarrow \Box^+$
(ii)	$\neg\Box^+\neg A \to \neg\Box\neg A$	Contraposing (i)
(iii)	$\Diamond^+ A \to \Diamond A$	(ii), Definition of $\Diamond$, $\Diamond^+$.

Something has gone wrong, then. The formula on line (iii) certainly isn't valid for all choices of A, as is most evident by taking A as $p \wedge \neg p$ or—what from now on we will assume available as a primitive 0-ary connective—$\bot$ (semantically interpreted as true at $x \in W$ iff $x = t$). For these choices of A, not only is the implication $\Diamond^+ A \to \Diamond A$ invalid, but it has an antecedent which is valid and a consequent which is not. Indeed perhaps it is obvious to the reader what exactly has gone wrong: the faulty transition was that from line (ii) to line (iii). All that has been said about $\Diamond$ and $\Diamond^+$ was said in the Sect. 1 and then applied in the Sect. 2, which amounted to extending our earlier truth-definition in the manner indicated below, in which the first column is blank, since we have already said that *every* formula—including those with $\Diamond$ or $\Diamond^+$ as main connective—is true at t in any model:

If $x = t$:	If $x \neq t$:
	$\mathcal{M} \models_x \Diamond A$ iff for some $y \in W \smallsetminus \{t\}$, $\mathcal{M} \models_y A$; $\mathcal{M} \models_x \Diamond^+ A$ iff for some $y \in W$, $\mathcal{M} \models_y A$.

We now have to see whether $\Diamond A$ and $\Diamond^+ A$ end up respectively, equivalent to $\neg\Box\neg A$ and $\neg\Box^+\neg A$, as the cavalier reference to definitions in the justification of line (iii) above would require. Let us take the case $\Diamond A$ first. Since for $\mathcal{M} = \langle W, t, V \rangle$ all formulas are true at t, $\Diamond A$ and $\neg\Box\neg A$ both take the same truth-value at t in $\mathcal{M}$, leaving the case of $x \neq t$ to consider. Calculating the conditions under which $\mathcal{M} \models_x \neg\Box\neg A$ for $x \neq t$, we find:

$\mathcal{M} \models_x \neg\Box\neg A$ iff $\mathcal{M} \not\models_x \Box\neg A$ (since $x \neq t$) iff for some $y \in W\smallsetminus\{t\}$, $\mathcal{M} \not\models_y \neg A$ iff (since $y \neq t$) $\mathcal{M} \models_y A$; in summary, then, $\mathcal{M} \models_x \neg\Box\neg A$ iff $y \in W\smallsetminus\{t\}$, $\mathcal{M} \models_y A$.

Since these are precisely the conditions under which $\mathcal{M} \models_x \Diamond A$, the equivalence in question holds unrestrictedly and we pass to the case of $\Box^+$ and $\Diamond^+$. For the same reason as before, the formulas $\Diamond^+ A$ and $\neg\Box^+\neg A$ must assume the same truth-value at t, and we are left to ponder when $\mathcal{M} \models_x \neg\Box^+\neg A$ for $x \neq t$:

$$\mathcal{M} \models_x \neg\Box^+\neg A \text{ iff } \mathcal{M} \not\models_x \Box^+\neg A \text{ (since } x \neq t) \text{ iff for some } y \in W, \mathcal{M} \not\models_y \neg A \ldots$$

We cannot continue to reason as before that this last conclusion holds iff $\mathcal{M} \models_y A$, since the "only if" direction here would require that $y \neq t$, which are not given. But, summarizing the reasoning to the stage at which we are blocked here, we have:

$$\mathcal{M} \models_x \neg\Box^+\neg A \text{ iff for some } y \in W, \mathcal{M} \not\models_y \neg A,$$

and noting that $\mathcal{M} \not\models_y \neg A$ iff $y \neq t$ and $\mathcal{M} \not\models_y \neg A$ (since if $y = t, \neg A$, like everything else, would be true at y) and so $\mathcal{M} \not\models_y \neg A$ iff $y \neq t$ and $\mathcal{M} \models_y A$. We conclude accordingly that the following two formulations:

$$\text{for some } y \in W, \mathcal{M} \not\models_y \neg A \quad \textit{and} \quad \text{for some } y \in W \smallsetminus \{t\}, \mathcal{M} \models_y A,$$

are equivalent. Since these are precisely the truth-conditions for $\Diamond A$ at x, we have shown that $\neg\Box^+\neg A$ is equivalent not to $\Diamond^+ A$ but to the weaker $\Diamond A$.[10] We have as valid the implication $\neg\Box^+\neg A \to \Diamond^+ A$ but not (in general) its converse. Indeed the failure of the latter principle is already evident from the fact that $\Diamond^+\neg(p \to p)$ is valid, as is $\Box^+(p \to p)$—so if the latter implied $\neg\Diamond^+\neg(p \to p)$, the set of valid formulas would be inconsistent, which is obviously not the case. Thus although $\Box^+$ and $\Diamond^+$ were initially explicated semantically as quantifying universally and existentially (respectively) over the universes of our models—an explication to which we shall return in the following section—they are not dual operators in the usual sense of modal logic, because of the peculiarities of one denizen of those universes, namely the element t thereof.[11]

[10] As we were already in a position to observe from the fact that $\Box$ and $\Box^+$ form equivalent formulas when prefixed to the same formula, together with the readily verified replacement property for the present logic: that if $A \leftrightarrow B$ is valid, so is $C \leftrightarrow D$, where C and D differ in that C has A as a subformula in one or more positions where D has B. (Here $A \leftrightarrow B$ can be treated as an abbreviation for $(A \to B) \wedge (B \to A)$; note that $A \leftrightarrow B$ is true throughout a model just in case A and B are true at precisely the same points in the model.) Again, we could have come at this observation—the failure of duality between $\Diamond^+$ and $\Box^+$—by another route: namely, the observation that the property of a 1-ary operator O, familiar from the case of contingency and noncontingency operators in modal logic, that the OA and $O\neg A$ are equivalent for any formula A, is possessed by $\Diamond^+$ but not by $\Box^+$, as it would have to be if these were related by 'circumnegation' duality.

[11] Suppose we worked instead with models $\langle W, u, V\rangle$ and a semantics like that presented above except that at the point u no formula is true—a single 'gap' element instead of a single 'glut' element—and we again interpreted $\Box$ ($\Diamond$) as quantifying universally (existentially) over $W \smallsetminus \{u\}$, and $\Box^+$ ($\Diamond^+$) as quantifying universally (existentially) over W. In this case it would be $\Diamond^+$ and $\Diamond$ (rather than $\Box^+$ and $\Box$) that were

Thus we find we are not after all dealing with a normal bimodal logic: $\Box$ and $\Diamond$ behave normally (and, more specifically, have $\mathsf{S5}$ for their logic), while (because of the failure of duality) $\Box^+$ and $\Diamond^+$ do not.[12] Ignoring this latter point would have give an even simpler derivation of triviality from $\Diamond^+$-modalized triviality than $(1)^+$–$(3)^+$, starting with $\Diamond^+\bot$, since normality would dictate that $\neg\Diamond^+\bot$ was provable and hence every other formula too. We return to the derivation $(1)^+$–$(3)^+$ in the following section.

Although not interdefinable in the expected way, the above discussion does show that each of $\Box^+$, $\Diamond^+$, is definable (in the logic comprising the set of all valid formulas) in terms of $\Box$ and the non-modal vocabulary, by:

$$\Box^+ A = \Box A \qquad\qquad \Diamond^+ A = A \to A.$$

Thus the set of valid formulas, although introduced above using the models $\langle W, t, V\rangle$ can be redescribed using just traditional—standard truth-definition, no t—simplified Kripke models (‘Carnap models’) $\langle W, V\rangle$ with the implicit accessibility relation (for $\Box, \Diamond$) being $W \times W$. We are dealing with nothing more than (propositional) $\mathsf{S5}$ enhanced by some unusual—and potentially confusing—notational features (“$\Box^+$”, “$\Diamond^+$”).[13] A complete semantic account of the logic is therefore, available in terms of models from which, despite its role in motivating this logic, the special element t

Footnote 11 continued
equivalent and $\Box^+$ would be a constant-false operator (rather than having $\Diamond^+$ as a constant-true operator). There would be the added complication as to how to define validity, since no formulas would be true throughout any model. One might instead require truth throughout $W \smallsetminus \{u\}$. A more systematic response would be to stop working within the framework of logics as sets of formulas and think instead of (generalized) consequence relations. Of course one could also have both t and u present at once, so that both aspects of impossibility distinguished in Note 1 as impossibility by commission and impossibility by omission would be in play simultaneously. And, somewhat further afield, one could explore the simultaneous presence of both features not only with respect to a single model but with respect to a single point in a model: see the references in Note 1.

12 What if we introduced a new operator $\Box^{\oplus}$ by the definition $\Box^{\oplus}A = \neg\Diamond^+\neg A$? $\Box^{\oplus}$ would not then qualify as a normal box operator, since $\neg\Box^{\oplus}A$, rather than $\Box^{\oplus}A$, would be valid for any valid formula A (indeed, for *any* formula A, since $\Box^{\oplus}$ would behave as a constant-false operator). The present duality issue complicates such remarks as the following, the opening sentence of Mortensen (1989): “I begin with the thesis of *possibilism*, by which I meand the group of theses that all propositions are possible, or possibly true, that all propositions are contingent, that no proposition is necessary.” If possibilism$^+$ by contrast, is defined as the thesis that every proposition (to use Mortensen’s word) is possible$^+$ (= true in some world), then as we have seen, this by no means implies that no proposition is is necessary$^+$ (true in all worlds)—not that Mortensen explicitly says his various theses are equivalent (though it would be hard to see that any one position called possibilism has been singled out if they aren’t). In any case, the prominence in Mortensen (1989) of fallibilist considerations suggests an epistemic notion of possibility is in play, showing at best that nothing is certain, rather than than nothing is necessary.

13 More precisely: the present logic is simply a definitional extension of normal monomodal $\mathsf{S5}$. (From the ‘constant-true’ behaviour of $\Diamond^+$ giving rise to its definability, we may alternatively note that it does behave as a normal modal operator, but despite its appearance, as a Box rather than a Diamond operator: the Box operator of what is often called the *Verum* system.) One may well react that this has made modalized trivialism itself a ‘trivial’ matter (in another sense): someone maintaining that it is possible$^+$ that A, for all A, may intend this as the endorsement of a metaphysical hypothesis, rather than as something to build into a propositional logic. Rather, a weaker logic should be used for exploring the consequences of this hypothesis, a logic relative to which $\Diamond^+A$ is not provably equivalent to (and fully interreplaceable with) $A \to A$. We explore this idea in the Appendix below.

is altogether absent. While technically correct, this observation does not offer much diagnostic insight into what has gone wrong with the derivation $(1)^+$–$(3)^+$ which appeared to spell disaster, and it would be good to see what goes wrong with the reasoning in its native habitat with t explicitly in play. Let us turn to that project.

5 Diagnosis

The derivation in question began at line $(1)^+$ with $\Diamond^+ p$ and called it the 'Modalized trivialism axiom'. All formulas of the form $\Diamond^+ A$ are indeed valid according to the semantics we have been considering, in view of the presence of the element t in all the models, so there is no faulting line $(2)^+$ either, which takes A as $\Box^+ q$. This is what we need, since what is under investigation is whether this leads to trivialism *tout court*—or, in logical terms, triviality or inconsistency. (Recall that we do not need to distinguish these two in the present context.) In line $(3)^+$ we pass to q from line $(2)^+$ with the justification being the general principle: $\Diamond^+\Box^+ A \rightarrow A$, and it is on this principle that attention must be focussed. The label B was bandied about in connection with this schema but we see now that this was a confusion, based on the assumption that $\Diamond^+ A$ was equivalent to $\neg\Box^+\neg A$. In fact this was described—again confusedly—as a 'dual form' of B, with the latter understood to as $A \rightarrow \Box^+\Diamond^+ A$. The latter principle is indeed valid (i.e., every formula instantiating this schema is valid) according to the present semantics, though for the not very interesting reason that the consequent $\Box^+\Diamond^+ A$ is already valid in its own right, and to not very much effect since it is the former principle we needed for the transition from $(2)^+$ to $(3)^+$. And this principle $\Diamond^+\Box^+ A \rightarrow A$ is evidently not valid, since it has a valid antecedent (all $\Diamond^+$-formulas being valid) and is therefore, false in a model at any point—of necessity, a point other than t—falsifying its consequent (for a suitable choice of A: and the present q will do just fine, with V appropriately specified). Some further diagnostic remarks on the derivation $(1)^+$–$(3)^+$ follow an excursus on a closely related line of reasoning, which can be extracted from Beall (2006, 2009), though presented here in the simplified setting in which there is only one impossible ('abnormal', as Beall says), world per model (namely t, as above), and also abstracted from the concerns of Beall (2006) with Priest's 'Characterization Principle' (in Priest 2005).

For this example we need a new ingredient in our models, intuitively to be thought of as representing the actual world amongst the various possible worlds, for which we use the notation "$w_@$" to match Beall's use of "@" for the corresponding sentence operator. Models now have the form $\langle W, t, w_@, V\rangle$ where now both t and $w_@$ are elements of W—and despite the previous reference to possible worlds we don't need to require $w_@ \neq t$ (as long as we *allow* this). There has been considerable debate (see Hanson 2006 for a review) as to whether validity in such a setting should consist in truth throughout W in every model of this form, or just in truth at the element $w_@$ in each such model. We need not take a stand on that issue. The usual clause in modal actuality logic, using models without t (or the like) but equipped with $w_@$ would run as follows:

$$\mathcal{M} \models_x @A \text{ if and only if } \mathcal{M} \models_{w_@} A,$$

for all formulas A.[14] Beall (2006) writes: "There seems to be a notion of actuality that is expressed by an entirely uniform actuality operator," and makes it clear that the above clause is what he has in mind as encapsulating this notion, which is described as *uniform* to convey the idea that this is to hold for all $x \in W$ even when W harbours impossible/abnormal elements, such as, in the present simple case, t, with models of the form $\langle W, t, w_@, V\rangle$.[15]

Parallelling the earlier $(1)^+$–$(3)^+$ we now consider:

$(4)^+$	$\Diamond^+ p$	Modalized trivialism axiom
$(5)^+$	$\Diamond^+ @q$	From $(4)^+$ by Uniform Substitution ($\Box^+ q$ for p)
$(6)^+$	$@q$	From $(5)^+$ using $\Diamond^+ @A \to @A$.

Clearly something has gone wrong here, since the formulas on lines $(4)^+$ and $(6)^+$ are respectively, valid and invalid. The principle appealed to in justification of the latter $\Diamond^+ @A \to @A$ is a reflection of the above 'uniform' treatment of @: for any given choice of A, this conditional must be true at an arbitrarily selected point x, since if the antecedent is true at x there must be some point in the model, even if it is only t, where $@A$ is true, whence by the uniform clause for @, A must be true at $w_@$, so $@A$ is true at x.[16] But we see immediately that this uniform treatment of @ is in tension with the status of t as the trivial world verifying all formulas regardless of what happens at other points in the model, without which feature we lose the only guarantee we had

[14] What about an operator O which similarly directs us to evaluate what follows at the point t (rather than at $w_@$), in the sense that we have $\mathcal{M} \models_x OA$ if and only if $\mathcal{M} \models_t A$? It does not take too much thought to see that we have already had such an O in play—namely $\Diamond^+$.

[15] The issue here is not the uniform treatment of formulas of a given form by means of some or other condition (inductive clause in the truth-definition), but rather their uniform treatment *by means of a particular condition*. After all—a point well made in Priest (2009)—a non-uniform treatment using a partition $\{W_1, \ldots, W_k\}$ of W with associated conditions $\Phi_i(x, A_1, \ldots, A_n)$ (for $i = 1, \ldots, k$) for the truth of #-compounds (# some n-ary connective) as follows (in which some formal metalinguistic notation appears in the interests of brevity):

$$\text{For all } x \in W_i : \mathcal{M} \models_x \#(A_1, \ldots, A_n) \Leftrightarrow \Phi_i(x, A_1, \ldots, A_n)$$

can be 'uniformized' to:

$$\text{For all } x \in W : \mathcal{M} \models_x \#(A_1, \ldots, A_n) \Leftrightarrow \bigvee_{i=1}^{k} (x \in W_i \wedge \Phi_i(x, A_1, \ldots, A_n)),$$

frequently with further simplification possible, as in the present case with the partition $\{W \setminus \{t\}, \{t\}\}$ (so $k = 2$): letting $\Phi_\#(x, A_1, \ldots, A_n)$ be the usual non-trivial condition associated with the connective #—i.e. that given in the second ("$x \neq t$") column of the earlier semantic tables—we have the uniform version: $\mathcal{M} \models_x \#(A_1, \ldots, A_n)$ iff $x = t$ or $\Phi_\#(x, A_1, \ldots, A_n)$. Further simplification is possible when $\Phi_\#(x, A_1, \ldots, A_n))$ does not depend on the A_j, as with the condition for $\perp$ (where the A_j are not even present) and we may take $\Phi_\perp(x)$ as $x \neq x$, so we were able to drop this disjunct entirely, giving the formulation used earlier: $\mathcal{M} \models_x \perp$ iff $x = t$. Alternatively, we may be able to drop the $x = t$ disjunct instead, as with $\top$, taking $\Phi_\top(x)$ as $x = x$. This is what happens with $\wedge$ and $\vee$ in what was called the "first strategy" regarding them in Sect. 3 (see Note 5), and with $\Diamond^+$ as noted in the final paragraph of this section.

[16] On the account of validity which identifies it with truth at the $w_@$-element of each model, we can cite the simpler $\Diamond^+ @A \to A$ as our justifying principle, and up with q itself (as opposed to $@q$) at line $(6)^+$.

that $\Diamond^+@q$, the formula at line $(5)^+$, was valid.[17] Priest (2009) expresses a similar sentiment in the following words:

> Quite generally, one is never at liberty to suppose that the truth conditions of an operator that involve a world-shift are uniform across all worlds. If one were, there would be formulas that held at *all* worlds, possible and impossible. But impossible worlds cannot be constrained in such a way: they can violate any constraint. Such, indeed, is the nature of the beasts.

Similar, but not quite the same. There is no problem for the present line of thought with there being formulas true at all worlds, possible or otherwise (in all models): these are the formulas we have been calling valid. This certainly does become a problem for ventures such as that described in Note 11, which is a simple special case of the general phenomenon (which Priest is concerned to accommodate) of impossibility through omission rather than commission, as it was put in Note 1. The passage just quoted from Priest (2009) is immediately followed by that given below, however, in which the issue is more precisely articulated, though in more informal terms and incorporating a reference to imaginability more pertinent to Priest's discussion than to ours:[18]

> In the context of intentionality, another way to see the point with respect to @ is as follows. For any meaningful claim, in particular $@A$, we can imagine that it is realised. If the truth conditions for @ allowed information to bleed back from an arbitrary world to the actual world, then we would be able to infer that A is actually true. Clearly this is not kosher: one can never move from the contents of one's imaginings to the real world—nice as this would be sometimes.

Let us apply this line of thought to the case of $(1)^+$–$(3)^+$. The first two lines of this derivation exploit the fact that models come equipped with the all-verifying world t: this is why all formulas of the form $\Diamond^+A$ are valid, including all such formulas in which A is $\Box^+q$. The justification for passing from this to q, namely $\Diamond^+\Box^+q \rightarrow q$ would have to be that even if $\Box^+q$ is only true at t, this is enough to allow us to conclude that q is true everywhere. As we have seen, this part of the reasoning would go equally well with $\Box q$ rather than $\Box^+q$, and with this replacement made, in terminology mirroring Beall's, we could say there "seems to be a notion of necessity that is expressed by an entirely uniform necessity operator." But again, this would conflict with the status of t as a trivial world, since whether or not $\Box A$ (or $\Box^+A$) is true at t would now depend on the truth of A at other points, and no longer be guaranteed come what may. Stipulating *both* that the truth of $\Box^+$ anywhere requires the truth of A everywhere, *and* that t verifies all formulas, would be a case of trying to have one's cake and eat it too: a choice must be made between these potentially conflicting

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[17] We could still retain the formula at line $(4)^+$, by insisting that $t \in V(p_i)$ for each propositional variable p_i, and make provision, as in the earlier semantics, for t to live up to its name ("trivial") for other @-free formulas, but uniform substitution would still fail for the case of @-formulas. To put it another way, we would lose the validity of $\Diamond^+A$ as a schema.

[18] The notation here has been adjusted to match Beall's; Priest himself uses "$\mathcal{A}$" rather than "@", and to avert potentially arising confusion chooses "B" rather than "A" as a schematic letter.

stipulations.[19] As in the "actually" case discussed in terms of unwanted bleed-back by Priest, one is "never at liberty to suppose that the truth conditions of an operator that involve a world-shift are uniform across all worlds", though in fact the point applies whether or not a world-shift is involved.[20] Beall might as well have said that there "seems to be a notion of negation that is expressed by an entirely uniform negation operator," meaning—and here we write the supposed operator as "$\sim$"—that for any $x \in W$, $\sim A$ should be true at $x \in W$ iff A is not true at $x \in W$ (relative to any given model). This would not be compatible with the project of having all formulas true at $t \in W$, since it conflicts, for any given formula B, with having $\sim B$ as well as B true at t. (Similarly, in introducing $\bot$ earlier, we did so with the semantic gloss that $\bot$ was to count as true at $x \in W$ iff $x = t$, rather than that $\bot$ is to count as true at no point whatever, since the latter would have been inconsistent with the presence, in each—indeed in any—model, of t as an all-verifying point.)

In the preceding section $\Box^+$ and $\Diamond^+$ were described as having been introduced initially with a semantic gloss to the effect they quantified universally and existentially (respectively) over the universes W of our models $\langle W, t, V\rangle$, but the semantics of the Sect. 3 reveals this to be an oversimplification, applying only to $\Box^+$ and $\Diamond^+$ as the main operators of formulas being evaluated at points in $W \setminus \{t\}$. For t itself, there was a once-and-for-all stipulation that every formula was to count as true. In the case of $\Diamond^+$, as it happens (see Note 15), the 'existential quantification' characterization remains correct nevertheless:

$$\langle W, t, V\rangle \models_t \Diamond^+ A \text{ if and only if for some } x \in W, \langle W, t, V\rangle \models_x A,$$

for the degenerate reason that regardless of the formula A, both sides are satisfied (in the case of the right-hand side, taking t itself as the required x).[21] But the analogous characterization in the case of $\Box^+$ (replacing $\Diamond^+$ here, with "all" replacing "some" on the right) would not be correct. Again the condition on the left is automatically satisfied, regardless of A, in view of the way t behaves, but the right-hand condition may not be satisfied. (e.g., take A as $\bot$ and any W with $W \setminus \{t\} \neq \varnothing$.) And in line $(2)^+$ of our earlier derivation, $\Diamond^+\Box^+ q$ provides a formula for whose truth at an arbitrary point is secured in the last-resort case ($\Box^+ q$ false everywhere else) by the truth of $\Box^+ q$ at t: but for the truth of $\Box^+ q$ at t to be guaranteed in this case, "$\Box^+$" must *not* be interpreted at t by the uniform 'true throughout W' condition, in the absence of which,

19 Well, there is one way of both having and eating it, but this makes the cake very small indeed. If the models were constrained so that $W \setminus \{t\} = \varnothing$, the conflict just alluded to could not arise. But this would not make room for much by way of an argument from modalized trivialism to trivialism *tout court*, as they would amount to starting with the latter: the logic determined by this class of models would simply be the set of all formulas.

20 Priest is well aware of this in connection with such cases as that of negation we are about to present in essentially the same way as he does in Priest (1990) and Chapter 5 of Priest (2006). This shows that the presence of a world-shifting clause in the truth-definition is not necessary for the current ('uniformity') problem to arise. The example of $\Diamond^+$, whether given the original clause involving existential quantification over W, or the alternative presented in Note 14, shows that it is not sufficient either.

21 Compare the characterization of $\Diamond^+$ given in Note 14.

as already remarked, we lose any justification for $\Diamond^+\Box^+ q \to q$, the other crucial step in the reasoning.

Appendix: A Weaker Logic

The derivation $(1)^+$–$(3)^+$ with which we have been concerned was a modal deduction of trivialism from modalized trivialism in the sense of starting from an axiom to the effect that everything was possibly$^+$ true and ending in the conclusion that everything was true. There is another way—foreshadowed in Note 13—of thinking of the claim that modalized trivialism entails trivialism, however, in which we derive $\perp$, say, from $\Diamond^+\perp$ as a mere hypothesis, rather than as an axiom; semantically, we will accordingly not want the latter formula to be valid, but we postpone describing the semantics for a moment. Since proof-theoretically we are still working within the axiomatic approach, this amounts to providing, not a proof of the latter as a theorem from the former as an axiom, but a proof of the implication having the latter as consequent and the former as antecedent. The derivation in question looks like this:

$(7)^+ \Diamond^+\perp \to \Diamond^+\Box^+\perp$	Justification given below.
$(8)^+ \Diamond^+\Box^+\perp \to \perp$	Instance of $\Diamond^+\Box^+ A \to A$
$(9)^+ \Diamond^+\perp \to \perp$	Truth-functional consequence of $(7)^+$ and $(8)^+$.

The semantic account we offer is of models in which there may be no trivial world(s); there is no advantage in having more than one such world but since it does no harm to allow this, we take the models to have a set T of trivial worlds, and—this is the crucial provision—allow this set to be empty. Thus models now take the form $\langle W, T, V\rangle$ in which W and V are as before and $T \subseteq W$. Truth is defined inductively for formulas relative to such a model at points in $W \setminus T$ as before (for points in $W \setminus \{t\}$)—and in particular $\Box^+ A$ and $\Box A$ are true at such points just in case A is true throughout W and throughout $W \setminus T$, respectively—while there is the outright stipulation that every formula is true at all $x \in T$.[22] As before, the valid formulas are those true at every point in every such model. Thus not all formulas of the form $\Diamond^+ A$ are valid, since, e.g., with $A = \perp$, such formulas will not be true at any point in $\langle W, T, V\rangle$ when $T = \varnothing$. However, the monotone rule for $\Diamond^+$:

$$\frac{A \to B}{\Diamond^+ A \to \Diamond^+ B}$$

is easily seen to preserve validity (indeed to preserve the property of being true throughout any given model). The justification of line $(7)^+$ above is that it results from applying this rule to the tautology $\perp \to \Box^+\perp$. The formula on line $(9)^+$, which we could write more familiarly as $\neg\Diamond^+\perp$, is by contrast not valid, being false at any point in $W \setminus T$ in a model $\langle W, T, V\rangle$ with $T \neq \varnothing$. And again (as in the case of $(1)^+$–$(3)^+$) the fault lies with the appeal to the pseudo-Brouwerian schema $\Diamond^+\Box^+ A \to A$ at line $(8)^+$, putatively the dual of $A \to \Box^+\Diamond^+ A$ which is, as before, valid (has only valid instances,

[22] Thus all earlier conditions—such as those in Note 15—are understood as having "$x = t$" ("$x \neq t$") replaced in them by "$x \in T$" ("$x \notin T$").

that is), though not, this time, simply because it has a valid consequent. $\Diamond^+\Box^+ A \to A$ fails to be valid in general for the same reason as before, which, transposed into the present setting, becomes: because of models in which T is non-empty. Thus the revised argument (7)$^+$–(9)$^+$ fares no better than the prototype (1)$^+$–(3)$^+$. The current logic raises some points of some interest in its own right, however, which we conclude by detailing.

Evidently, the set of formulas valid on the present semantics is strictly included in that valid on the earlier semantics with models $\langle W, t, V\rangle$, since the latter amounts to restricting attention to the current models $\langle W, T, V\rangle$ with $T \neq \varnothing$. From this consideration the above observation that $\neg\Diamond^+\bot$ is invalid on the present semantics is immediate: if it were, then *a fortiori* it would be valid on the earlier semantics, whereas it is the negation of a formula valid on that semantics. On the contrary, the current logic (conceived of as the set of formulas valid on the current semantics) is agnostic, or non-committal, on the question of whether there is a trivial world, as is witnessed by its containing the Halldén-unreasonable disjunction $\neg\Diamond^+\bot \vee \Diamond^+\bot$.[23]

Some modally unusual aspects of the stronger logic considered earlier are still present. It is not hard to see, for example, that, while $\Diamond A$ and $\Diamond^+ A$ are not equivalent in this weaker setting, $\Box A$ and $\Box^+ A$ are still equivalent, so we can dispense with $\Box^+$ and simply use $\Box$, as before. On the other hand, $\Diamond^+$ no longer behaves as a constant-true operator (since we allow $T = \varnothing$ in our models[24]) and we can see that it is no longer definable in terms of the remaining primitives by considering models $\mathcal{M}_0 = \langle W_0, T_0, V_0\rangle$ and $\mathcal{M}_1 = \langle W_1, T_1, V_1\rangle$ with $W_0 = \{w\}$, $W_1 = \{w, t\}$ $(w \neq t)$, $T_0 = \varnothing$, $T_1 = \{t\}$, $V_0(p_i) = W_0$, $V_1(p_i) = W_1$ (for each propositional variable p_i). Then one checks by induction on the construction of A that for all $\Diamond^+$-free formulas A, we have:

$$\mathcal{M}_0 \models_w A \Leftrightarrow \mathcal{M}_1 \models_w A$$

even though $\mathcal{M}_0 \not\models_w \Diamond^+\bot$ while $\mathcal{M}_1 \models_w \Diamond^+\bot$, showing that $\Diamond^+\bot$ is not equivalent to any $\Diamond^+$-free formulas, and therefore, that $\Diamond^+$ is not definable.[25]

The undefinability of $\Diamond^+$ gives rise to the question of how to axiomatize the weaker logic, from whose language we assumed $\Box^+$ excised (in favour of $\Box$) for simplicity, and one answer to that question is that we take any basis for monomodal $\mathsf{S5}$ for $\Box$ and $\Diamond$, which includes the rules of Modus Ponens and ($\Box$-)Necessitation, and add the above monotone rule for $\Diamond^+$ along with all formulas instantiating the following

[23] For the connection between Halldén incompleteness and agnosticism, see Humberstone (2007) and references there given.

[24] For this reason we also lose the equivalence of $\Diamond^+ A$ with $\Diamond^+\neg A$; the operator $\Box^\oplus$ as defined in Note 12 is still not a normal Box operator, $\Box^\oplus\top$ being invalid, for instance. Nor can $\Diamond^+$ itself be taken as a normal Box operator, since for example, $(\Diamond^+ p \wedge \Diamond^+ q) \to \Diamond^+(p \wedge q)$ is not valid.

[25] If one is interested in minimizing the arity of one's primitive connectives, however, one might observe that adding the sentential constant (nullary connective) **c** true at any point in a model $\langle W, T, V\rangle$ just in case $T \neq \varnothing$—so that **c** behaves like $\Diamond^+\bot$—then $\Diamond^+ A$ can be defined as $\mathbf{c} \vee \Diamond A$.

schemata as axioms:[26]

$$\Diamond A \to \Diamond^{+} A \qquad \Diamond^{+} A \to (\Diamond A \vee \Diamond^{+}\bot) \qquad \Diamond^{+} A \to \Box\Diamond^{+} A.$$

Note incidentally that the second schema has, when A is taken as p, an instance which can be re-bracketed to yield another example of a Halldén-unreasonable disjunction for this logic: $(\Diamond^{+} p \to \Diamond p) \vee \Diamond^{+}\bot$ (in which the $\bot$ can be replaced by q, if desired). As with the earlier example of such a disjunction, adding the first disjunct as a new axiom would give a version of $\mathsf{S5}$ with the notational redundancy of having two (equivalent) possibility operators $\Diamond$ and $\Diamond^{+}$, while adding the second returns us to the logic of the main body of our discussion: the two options the present logic leaves open as to the existence of trivial worlds. It is a pleasant exercise to derive further formulas valid on the present semantics from the basis described here (such as $p \to \Diamond^{+} p$, $\neg\Diamond^{+} p \to \Box\neg\Diamond^{+} p$, and $\Diamond^{+}(p \vee q) \to (\Diamond^{+} p \vee \Diamond^{+} q)$), and in fact all valid formulas are similarly derivable. A simple Hintikka-style variation on the Scott–Makinson canonical model method of proof is given in the following paragraph.

For completeness, take an unprovable formula A and extend $\{\neg A\}$ to a set of formulas—call it w—maximally consistent w.r.t. the present logic. We construct a model $\mathcal{M} = \langle W, T, V\rangle$ by taking for $W \setminus T$ the set of all maximal consistent sets of formulas which contain precisely the same $\Box$-formulas as w does. If w contains $\Diamond^{+}\bot$, then $T = \{t\}$ where t is the set of all formulas; otherwise $T = \varnothing$. (Either way, note that $w \notin T$.) As usual $V(p_i) = \{x \in W \mid p_i \in x\}$. The only novelty now in showing that for any formula B, $\mathcal{M} \models_x B$ iff $B \in x$—which will allow us to conclude that for our unprovable formula A, $\mathcal{M} \not\models_w A$ (and hence that A is invalid)—arises for B of the form $\Diamond^{+} C$, so let us sketch the reasoning for that case. Suppose that $\Diamond^{+} C \in x \in W$. By the second axiom(-schema) inset above we conclude that either $\Diamond C \in x$ or $\Diamond^{+}\bot \in x$. In the former case, taking it that truth and membership have already been shown to coincide for $\Diamond$-formulas, we have some $y \in W \setminus T$ with $C \in y$, so we certainly have some $y \in W$ with $C \in y$, and thus by the inductive hypothesis $\mathcal{M} \models_y C$, showing, as required, that $\mathcal{M} \models_x \Diamond C$. In the latter case, with $\Diamond^{+}\bot \in x$, we have $\Box\Diamond^{+}\bot \in x$ by the third axiom schema, and so, by the way W was defined $\Box\Diamond^{+}\bot \in w$, in which case (by $\mathsf{S5}$ reasoning) $\Diamond^{+}\bot \in w$, so $T = \{t\}$ and t will do as an element of W containing—and hence by the inductive hypothesis, verifying—the formula C. Thus we have shown that $\Diamond^{+} C \in x \in W \Rightarrow \mathcal{M} \models_x \Diamond^{+} C$. For the converse, suppose that $\mathcal{M} \models_x \Diamond^{+} C$, i.e., for some $y \in W$, $\mathcal{M} \models_y C$, which is to say, by the inductive hypothesis: for some $y \in W$, $C \in y$. If $y \notin T$, then $\Diamond C \in x$ and we get $\Diamond^{+} C \in x$ by the first axiom listed above. If $y \in T$ then $y = t$ so $\Diamond^{+}\bot \in w$, so $\Diamond^{+}\bot \in x$ (reasoning as before, since $\Box\Diamond^{+}\bot \in w$, so $\Box\Diamond^{+}\bot \in x$, so $\Diamond^{+}\bot \in x$). Applying the monotone rule for $\Diamond^{+}$ to the implication $\bot \to C$ means that $\Diamond^{+}\bot \to \Diamond^{+} C$ is provable, so since $\Diamond^{+}\bot \in x$, we have $\Diamond^{+} C \in x$, as desired, completing the proof.

Since the only appeal to the monotone rule for $\Diamond^{+}$ in the above completeness proof was to the special case with $\bot$ as antecedent, it follows that we could replace the rule

[26] We could, as in the earlier discussion, equally well use a single representative instance of each schema and the rule of Uniform Substitution to obtain the rest. If the $\mathsf{S5}$ part of the account, $\Box$ and $\Diamond$, is given by schemata, these must be understood as having instances in the extended language with $\Diamond^{+}$ present.

with the schema $\Diamond^+\bot \to \Diamond^+ A$. We close by showing how to derive the monotone rule from this new schema and the rest of the above axiomatic basis. Start with $A \to B$, *ex hypothesi* provable. Using the S5-derivable monotone rule for $\Diamond$ itself, we have $\Diamond A \to \Diamond B$, and hence, by appeal to the first axiom schema listed earlier, we get $(*)$: $\Diamond A \to \Diamond^+ B$. The second schema gives:

$$\Diamond^+ A \to (\Diamond A \vee \Diamond^+\bot),$$

so by truth-functional logic and $(*)$ we have $\Diamond^+ A \to (\Diamond^+ B \vee \Diamond^+\bot)$, and, now working on the second disjunct of the consequent and appealing to the new schema, we derive: $\Diamond^+ A \to (\Diamond^+ B \vee \Diamond^+ B)$, which we simplify to obtain the conclusion, $\Diamond^+ A \to \Diamond^+ B$, of the monotone rule for $\Diamond^+$.

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